

[illegible]

1. M-spike shield wall (infection)

WHAT YOU SEE

A giant red 'M' shield wall under an 'INFECTION ZONE' banner, ringed by bacteria (S. pneumoniae, E. coli, Klebsiella).

WHAT IT ENCODES

Infection is the leading cause of death in multiple myeloma. A monoclonal M-protein crowds out normal polyclonal immunoglobulin, producing functional hypogammaglobulinemia and impaired humoral immunity. Patients are especially prone to encapsulated organisms — S. pneumoniae and other respiratory/urinary pathogens.

BOARD TRAP

Despite a high total protein/globulin, myeloma patients are immunodeficient — the M-protein is useless; do not be reassured by a 'normal' total immunoglobulin level.

2. Claw B — Bone smashing

WHAT YOU SEE

A crab claw labeled 'Claw B — BONE' smashing a building, with 'CALCIUM' boulders and punched-out lesions.

WHAT IT ENCODES

The 'B' of CRAB is Bone lesions. Myeloma cells secrete RANKL and DKK1, activating osteoclasts and suppressing osteoblasts, producing purely lytic 'punched-out' lesions and pathologic fractures. Skeletal survey/whole-body low-dose CT or MRI/PET is used — classic bone scan is INSENSITIVE because there is no osteoblastic reaction.

BOARD TRAP

Bone (technetium) scan is falsely negative in myeloma because the lesions are osteoclastic/lytic with no osteoblast uptake — use skeletal survey, low-dose whole-body CT, or PET/MRI instead.

3. Bone pain / collapsed spine

WHAT YOU SEE

A collapsed spine and cord compression flag beside a 'BONE PAIN' label.

WHAT IT ENCODES

Bone pain (back/ribs) is the most common presenting symptom. Vertebral compression fractures cause height loss and can produce spinal cord compression, an oncologic emergency requiring urgent MRI and dexamethasone plus radiation/surgery.

BOARD TRAP

New focal back pain with neurologic signs in a myeloma patient is cord compression until proven otherwise — image emergently rather than attributing it to benign mechanical pain.

4. Claw A — Anemia / marrow

WHAT YOU SEE

A claw labeled 'Claw A — ANEMIA' over a 'MARROW' factory with EPO and reduced RBC output.

WHAT IT ENCODES

The 'A' of CRAB is Anemia, typically normocytic-normochromic from marrow replacement by plasma cells and relative EPO deficiency. Rouleaux formation on smear reflects the high paraprotein. Anemia plus a high total protein/globulin gap is a classic exam trigger to check SPEP.

BOARD TRAP

An elevated ESR with anemia and rouleaux points to a paraprotein — order SPEP/UPEP and free light chains, not just an iron panel.

5. Claw C — Calcium >11

WHAT YOU SEE

A claw labeled 'Claw C — CALCIUM' with a '>11 mg/dL' tag and 'stones, bones, groans, moans' icons.

WHAT IT ENCODES

The 'C' of CRAB is hyperCalcemia (corrected Ca >11 mg/dL or >1 mg/dL above ULN) from osteoclastic bone resorption. Symptoms: polyuria, constipation, confusion, nausea. Treat with aggressive IV normal saline first, then bisphosphonate (zoledronate) and calcitonin for rapid effect.

BOARD TRAP

First-line treatment of myeloma hypercalcemia is IV normal saline volume expansion — bisphosphonates work over days; loop diuretics are NOT first-line and only after euvolemia.

6. Claw R — Renal / cast nephropathy

WHAT YOU SEE

A clogged drain labeled 'Claw R — RENAL', with light chains, Tamm-Horsfall protein and 'CAST NEPHROPATHY'.

WHAT IT ENCODES

The 'R' of CRAB is Renal failure. The dominant mechanism is light-chain (cast) nephropathy: filtered free light chains bind Tamm-Horsfall protein in the distal tubule forming obstructing casts. Hypercalcemia and hyperuricemia add insult. Management: hydration, treat hypercalcemia, and rapid anti-myeloma therapy to lower light chains.

BOARD TRAP

The classic renal lesion of myeloma is cast nephropathy from FREE LIGHT CHAINS, not the intact M-protein — and AL amyloidosis/MIDD are separate light-chain renal lesions.

7. Dipstick negative for protein

WHAT YOU SEE

A detective with a urine dipstick reading 'DIPSTICK NEGATIVE' beside a 'FULL' UPEP cup.

WHAT IT ENCODES

Urine dipstick detects ALBUMIN but NOT immunoglobulin free light chains (Bence Jones protein). A negative dipstick with heavy proteinuria on 24-hour collection or UPEP/immunofixation signals light-chain (Bence Jones) proteinuria — the hallmark of myeloma cast nephropathy.

BOARD TRAP

A negative urine dipstick does NOT exclude myeloma proteinuria — dipstick misses light chains; you must order UPEP/urine immunofixation or 24-hour protein.

8. AL amyloidosis (leaking glucose)

WHAT YOU SEE

An 'AL AMYLOIDOSIS' label with 'LEAKING GLUCOSE' and tubular injury icon.

WHAT IT ENCODES

AL amyloidosis from misfolded light chains deposits in kidney (nephrotic-range albuminuria), heart (restrictive cardiomyopathy, low-voltage ECG), tongue (macroglossia), and nerves. Apple-green birefringence on Congo red stain confirms it. Proximal tubular light-chain injury can produce Fanconi syndrome with glucosuria.

BOARD TRAP

Nephrotic-range ALBUMINURIA in a plasma-cell disorder = AL amyloidosis (glomerular), distinct from cast nephropathy (tubular, light-chain) which spares albumin.

9. Hyperviscosity station

WHAT YOU SEE

A 'HYPERVISCOSITY STATION' with thick pipes, 'clear viscosity' and a plasmapheresis machine.

WHAT IT ENCODES

Hyperviscosity syndrome (blurred vision, mucosal bleeding, headache, confusion) is far more common with IgM (Waldenstrom) but can occur with high IgA/IgG myeloma. Funduscopy shows dilated 'sausage-link' retinal veins. Emergent treatment is plasmapheresis to remove the paraprotein, then treat the clone.

BOARD TRAP

Hyperviscosity is classically IgM (Waldenstrom macroglobulinemia), not typical IgG myeloma — and the emergency treatment is plasmapheresis, not transfusion (which worsens viscosity).

10. M-component / clotting hazard

WHAT YOU SEE

A 'CLOTTING HAZARD' panel listing antibody-coated, non-functioning clotting factors and 'M component'.

WHAT IT ENCODES

The paraprotein can coat platelets and clotting factors causing acquired bleeding (acquired von Willebrand, factor inhibition) — yet myeloma and its IMiD therapy (lenalidomide/thalidomide) also markedly raise VTE risk, so thromboprophylaxis (aspirin or anticoagulation) is standard with IMiD regimens.

BOARD TRAP

IMiDs (lenalidomide/thalidomide) carry a high VTE risk requiring prophylaxis — this is a frequently tested management point distinct from the paraprotein's bleeding effect.

11. Neurologic — weak legs / confused

WHAT YOU SEE

A 'NEUROLOGIC' panel with 'numb', 'confused', 'weak legs' figures.

WHAT IT ENCODES

Neurologic features arise from cord compression (weak legs, sensory level), hypercalcemia (confusion), hyperviscosity (CNS), amyloid/light-chain peripheral neuropathy, and POEMS. Bortezomib and thalidomide themselves cause peripheral neuropathy — a key dose-limiting toxicity.

BOARD TRAP

Sensory peripheral neuropathy in a treated myeloma patient is often drug-induced (bortezomib > thalidomide); switching to subcutaneous/weekly bortezomib reduces it.

12. Pseudohyponatremia trap

WHAT YOU SEE

A 'PSEUDOHYPONATREMIA TRAP' with a 'LOW Na+' chemistry slip and a 'PSEUDO — NO SALT!' sign.

WHAT IT ENCODES

A very high M-protein increases the plasma's non-aqueous (protein) volume, so older indirect ion-selective/flame-photometry sodium assays read a falsely LOW sodium — pseudohyponatremia. Measured serum osmolality is normal; direct ISE on a blood gas gives the true sodium.

BOARD TRAP

A 'low' sodium with normal measured osmolality and a huge paraprotein = pseudohyponatremia — do NOT treat with hypertonic saline; confirm with direct ISE (blood gas) sodium.

13. Calcium boulders / lytic skull

WHAT YOU SEE

Calcium 'boulders' tumbling from a smashed building with punched-out 'lytic lesions' lettering.

WHAT IT ENCODES

Punched-out lytic lesions are most numerous in the skull (pepper-pot/raindrop skull), vertebrae, ribs, and proximal long bones. They reflect uncoupled bone turnover; alkaline phosphatase is often NORMAL because there is no osteoblastic repair.

BOARD TRAP

A normal alkaline phosphatase despite extensive lytic bone disease is characteristic of myeloma (no osteoblastic response) and distinguishes it from osteoblastic metastases.

14. >11 mg/dL calcium flag

WHAT YOU SEE

A red flag reading '>11 mg/dL' planted near the calcium claw.

WHAT IT ENCODES

The CRAB calcium criterion is serum corrected calcium >11 mg/dL (or >1 mg/dL above the upper limit). Meeting any single CRAB feature (plus a clonal plasma-cell disorder) defines symptomatic myeloma requiring treatment, separating it from smoldering myeloma/MGUS.

BOARD TRAP

Any one CRAB criterion (or a myeloma-defining biomarker: ≥60% plasma cells, FLC ratio ≥100, >1 focal MRI lesion) upgrades smoldering to active myeloma needing treatment — you do NOT need all of CRAB.

15. Cloggin' drain (light-chain casts)

WHAT YOU SEE

A clogged sink pipe labeled 'CLOGGIN'" with light chains and Tamm-Horsfall protein forming a plug.

WHAT IT ENCODES

Mnemonic for the renal mechanism: free light chains + Tamm-Horsfall protein 'clog' the distal tubule. Avoiding nephrotoxins, NSAIDs, and IV contrast, plus volume repletion and rapid light-chain reduction (often bortezomib-based), preserves renal function.

BOARD TRAP

Avoid IV contrast and NSAIDs in myeloma renal disease — contrast can precipitate cast nephropathy/AKI in the setting of high light-chain load.
